Stroke Risk Profiles in a Subset of the Framingham Heart Study Cohort

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Introduction

The Framingham Heart Study is a prospective long term cohort study investigating cardiovascular disease risk which began in 1948. We investigate profiles associated with increased probability of risk of stroke in a subset of this study's population recorded from 1956 to approximately 1968. We are interested in whether age, systolic blood pressure, diabetes, smoking status, coronary heart disease, BMI, total cholesterol, and blood pressure medication use are risk factors for stroke. We will test whether these variables are significantly associated with stroke risk, with primary variables of interest being age, systolic blood pressure, and diabetes. Change in risk factors over the study duration is explored to better understand future directions using time-varying covariates.

Methods

Data comes from a subset of the Framingham Heart Study, with 4,434 individuals followed for 3 time periods, approximately 6 years apart, ranging from 1956 to 1968. Variables of interest included in the data are time until stroke or censorship, the stroke event, sex, age, diabetes (Y/N), systolic blood pressure, BMI, blood pressure medication use (Y/N), total cholesterol level, current smoking status (Y/N), and coronary heart disease (Y/N). For the risk factor analysis, we excluded individuals who had already had a stroke before the beginning of the study (TIMESTRK=0), only considered period 1, in order to capture baseline characteristics, and censored all individuals at 10 years (3652 days), resulting in 4,402 remaining individuals. We disregarded other events measured in the data. We also considered complete case analysis, removing individuals with missing information on the covariates we are interested in, and stratified the data into a male group and female group, with final sample sizes of 1,897 and 2,378 respectively. Missingness was assessed for meaningful patterns. None were found so data was assumed to be missing completely at random. To analyze change in risk factors, we used data on the same individuals for periods 1 and 3 for consistent interpretation.

To examine which risk profiles increase the probability of stroke in 10 years, we first visualized the survival curves using Kaplan Meier for different levels of the risk factors (stratifying continuous variables by quantile), showing that there are visible differences in the survival curves and that the risk factors are worth investigating. We are primarily interested in studying age, high systolic blood pressure (>160 mmHg) and diabetes as risk factors, but also want to identify whether blood pressure medication use, BMI, cholesterol levels, smoking status (current vs. not) and coronary heart disease also significantly contribute to the probability

Table 1: Participant Characteristics

	Male	**Female**
Characteristic	**N = 1,897**[1]	**N = 2,378**[1]
Age	50 (9)	50 (9)
Systolic Blood Pressure (mmHg)	132 (19)	134 (24)
Diabetes	56 (3.0%)	57 (2.4%)
Current Smoking / Tobacco Use (Y/N)	1,147 (60%)	958 (40%)
BMI	26.2 (3.4)	25.6 (4.5)
Blood Pressre Medication Use (Y/N)	40 (2.1%)	92 (3.9%)
Prevalent Coronary Heart Disease	113 (6.0%)	64 (2.7%)
Total Cholesterol Level	234 (42)	240 (46)
Stroke Event	48 (2.5%)	57 (2.4%)

[1]Mean (SD); n (%)

of stroke. To identify significant risk factors and compute their effects on the probability of stroke, we created Cox PH models for the female group and the male group, selecting risk factors to include using backward stepwise selection with the Akaike Information Criterion (AIC). Because correlation matrices showed moderate correlation between some variables, (e.g. systolic blood pressure and age having positive correlation of .49), multicollinearity was assessed in both final models by checking the variance inflation factors, a measure of how much the variance and significance of a variable can be affected by its correlation with another variable. Cox PH coefficients were assessed for significance using $\alpha = .05$. Adherence to Cox PH assumptions was assessed using Schoenfeld residual testing and visualization (plots available in appendix.) We allowed the final models for males and females to differ in order to be able to report probabilities calculated from the most parsimonious model available for both groups. The final model for males included age, systolic blood pressure, diabetes, and smoking. The final model for females included age, systolic blood pressure, and diabetes.

For both sex groups, and for ages 40, 50, and 60, we were primarily interested in the stroke

risk for the following profiles: someone with diabetes, someone with high blood pressure (we used 160 to compute our risk profiles), and someone with both. Additionally, because smoking was found to be predictive in the male group, we also considered the risk profile of a male smoker, as well as a male smoker with diabetes and high blood pressure. Probabilities for these risk profiles were computed using the final Cox PH models for each age group.

We also measured the change in each of the 8 risk factors listed above in the subset of our final study population that had data measured at both period 1 and period 3 (1,102 males and 1,438 females.) This subset was used to maintain consistency of the population we are drawing inference on throughout the analysis. Code for the analysis is available at https://github.com/jkebe2002/BIOS6624/tree/9c435c37471509d45ecdb37552575dbf7bd10cd7/P3 and the data is stored locally on Jacob Kebe's personal computer at C:/Users/jkebe/Downloads/frmgham2.csv.

Results

For males, the model with the lowest AIC included age, systolic blood pressure, diabetes, and smoking status, suggesting that this model fits the data better and includes more predictive variables (Portet, 2020) than other models that included age, systolic blood pressure, and diabetes as the null model. For females, the model with the lowest AIC was the null model. Among the VIFs for these two models, there was never one larger than 1.25, so multicollinearity was dismissed as an issue that could skew estimates and significance. Schoenfeld plots and tests showed no significant pattern signifying Cox PH model assumption violations.

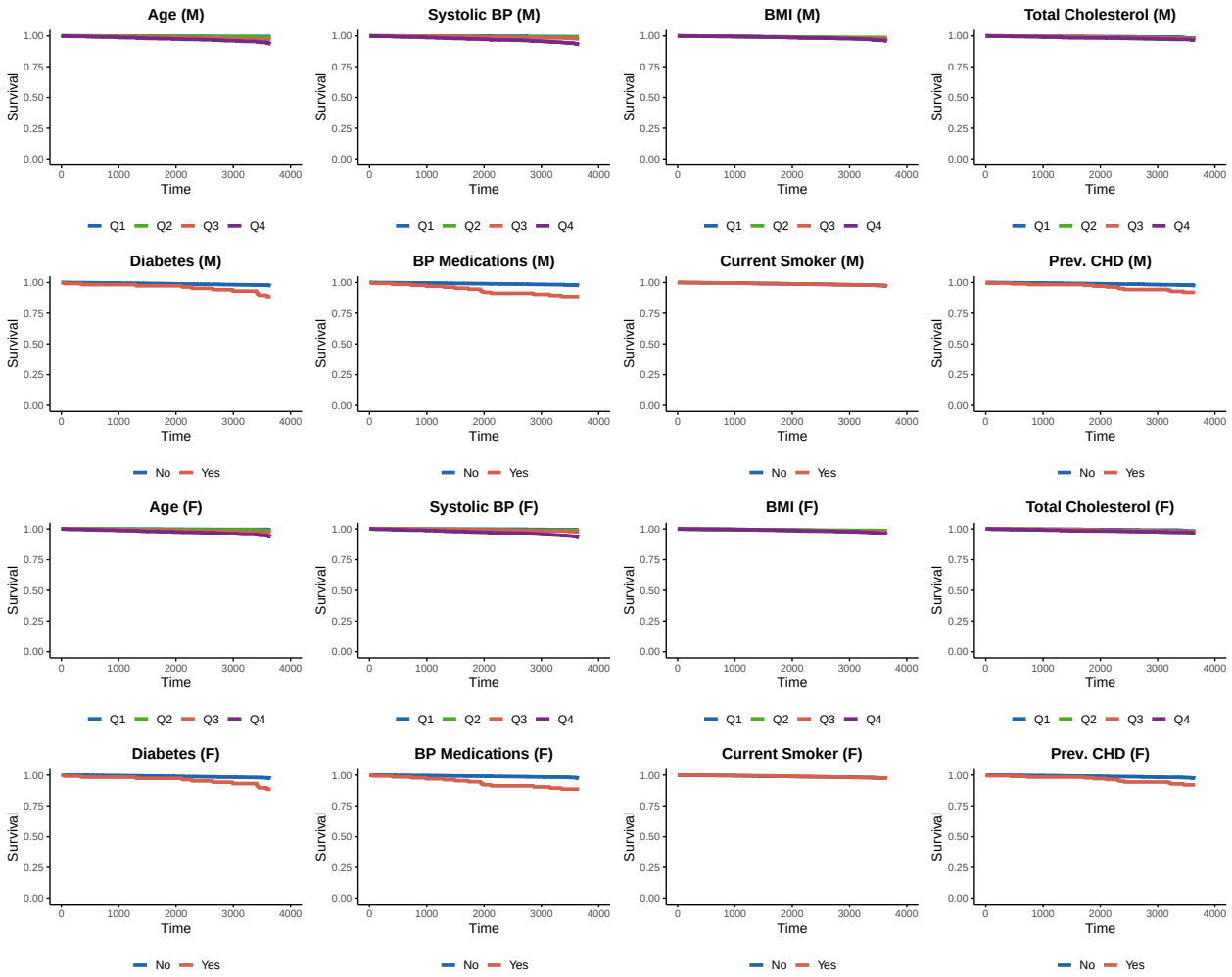

Figure 1: Kaplan Meier Curves for Potential Risk Factors

For males, every ten year increase in age is associated with 90.0% (83.1%, 97.3%) increased stroke risk when adjusting for systolic blood pressure, diabetes, and smoking status ($p < .0001$). Every ten unit increase in systolic blood pressure in males contributes to a 39.7% (95% CI: [38.2%,41.3%]) increased stroke risk, adjusting for age, diabetes, and smoking status ($p < .0001$). Males with diabetes have a risk of stroke in 10 years that is 380.7% (95% CI: [109.9%, 1000.5%]) higher than those without diabetes ($p = .0002$) when adjusting for age, systolic blood pressure, and smoking status. Males who smoke have a risk of stroke in 10 years that is 92.4% (95% CI: [3.6%, 257.3%]) higher than those who do not smoke (p =

.038) when adjuysting for age, systolic blood pressure, and diabetes. At $\alpha = .05$ These four variables were all found to be significant risk factors contributing to risk of stroke.

For females, every ten year increase in age is associated with a stroke risk increase of 103.2% (95% CI: [38.38%,198.5%]) when adjusting for systolic blood pressure and diabetes (p<.0001). For females, every 10 mmHg increase in systolic blood pressure contributes to a 29.3% increase (95% CI: [19.3%, 40.2%]) in risk of stroke when adjusting for age and diabetes (p< .0001). At $\alpha = .05$, Both age and systolic blood pressure were found to be factors significantly associated with risk of stroke in 10 years for the female group. Diabetes, although included in the null model for the female group, was not found to significantly alter stroke risk: Those with diabetes had a risk of stroke that was 176.7% (95% CI: [63.6%, 490.4%]) the risk of stroke in those without diabetes when adjusting for age and systolic blood pressure, providing no evidence of increased or decreased risk (p=.275).

Table 2 shows the probabilities of stroke in a ten year period for different risk profiles described in the methods, along with probabilities for individuals with median age and systolic blood pressure with no diabetes (and for men, nonsmokers). We see that as levels of risk factors increase, the probability of stroke increases, in concurrence with the hazard ratio estimates.

In the subset of this population with data for periods 1 and 3, systolic blood pressure increased, and trends for men and women differed for BMI and total cholesterol. Many people changed smoking status (22% of men and 11% of women stopped smoking, while 2.5% of men and 1.4% of women reported new smoking behavior). Also, 10% of men and 16% of women started taking blood pressure medication and .3% and .5% stopped, respectively. 12% of men and 6.3% of women were diagnosed with CHD after baseline, and 6.9% and 5.6%

Table 2: Probability of risk in a 10 year period for men and women in a sample from the Framingham Heart Study, using median age and systolic blood pressure, no diabetes, and for men, no smoking behavior, as the reference levels.

Risk Group	Males Prob. of Stroke @ 10yrs (95% CI)	Females Prob. (95% CI)
Median Age		
No Risk Factors	0.90% (0.33%, 1.46%)	1.13% (0.66%, 1.60%)
Age = 60		
High BP	5.11% (2.44%, 7.71%)	5.44% (3.87%, 6.97%)
Diabetes	22.27% (5.37%, 36.15%)	4.30% (0.00%, 8.62%)
Smoker	3.45% (1.72%, 5.16%)	–
High BP, Diabetes	22.27% (5.37%, 36.15%)	9.40% (0.10%, 17.84%)
High BP, Diabetes, Smoker	38.42% (7.06%, 59.20%)	–
Age = 50		
High BP	2.72% (1.10%, 4.31%)	2.71% (1.69%, 3.73%)
Diabetes	4.52% (0.60%, 8.28%)	2.14% (0.00%, 4.34%)
Smoker	1.83% (1.05%, 2.61%)	–
High BP, Diabetes	12.42% (1.80%, 21.88%)	4.74% (0.00%, 9.39%)
High BP, Diabetes, Smoker	22.52% (2.87%, 38.20%)	–
Age = 40		
High BP	1.44% (0.22%, 2.64%)	1.34% (0.39%, 2.28%)
Diabetes	2.40% (0.00%, 4.85%)	1.06% (0.00%, 2.30%)
Smoker	0.97% (0.36%, 1.57%)	–
High BP, Diabetes	6.74% (0.00%, 13.49%)	2.36% (0.00%, 5.12%)
High BP, Diabetes, Smoker	12.56% (0.00%, 24.19%)	–

were diagnosed with diabetes. Note that this is a biased sample as it excludes those without measurements for period 3. Individuals excluded may have dropped out for health-related reasons or may have died, so the subset may be healthier than the actual population.

Discussion

Over a ten year period, increases in age and systolic blood pressure, as well as presence of diabetes and smoking behavior, significantly increase the risk of stroke in men. In women,

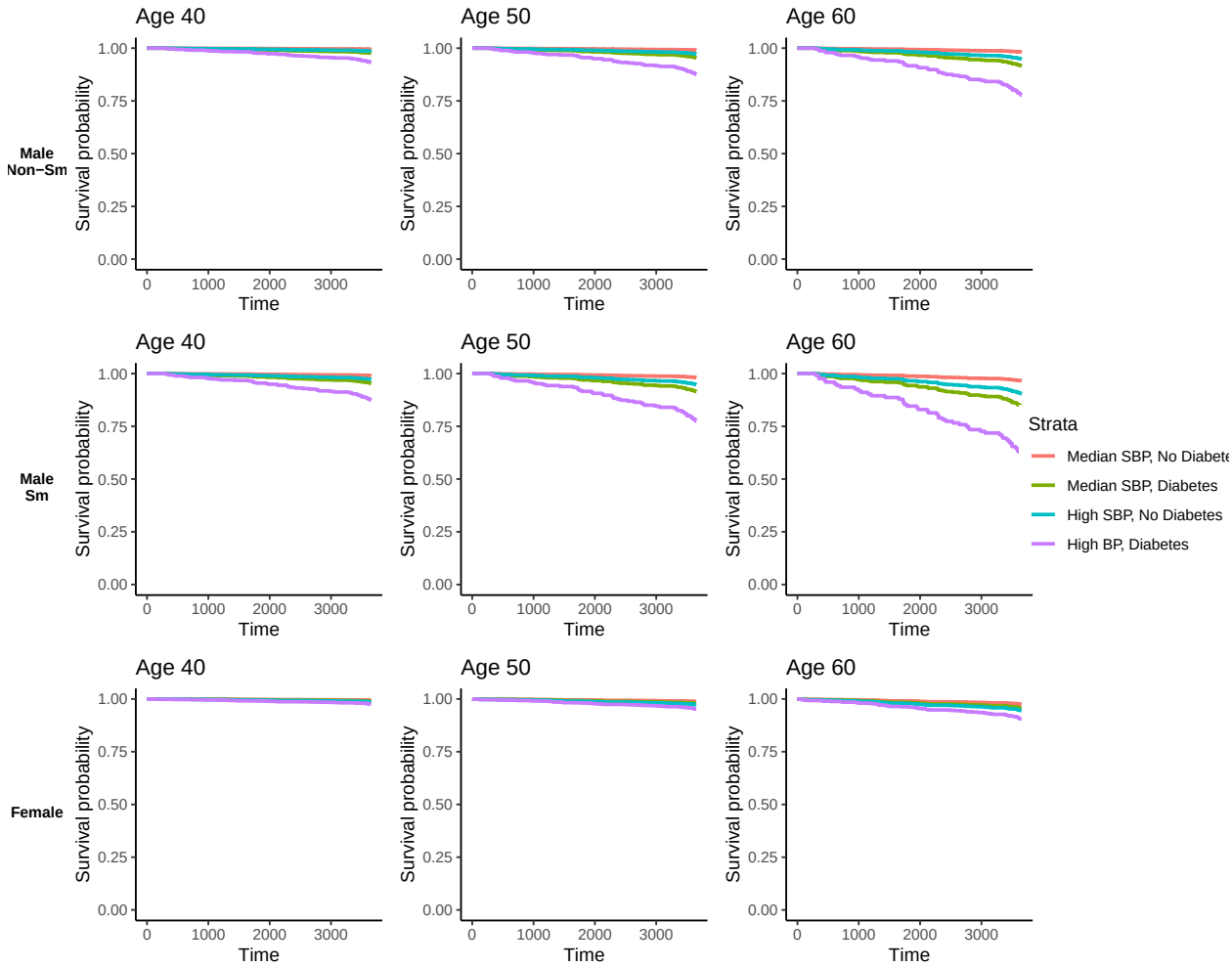

Figure 2: Survival curves by age, sex, and smoking status

increases in age and systolic blood pressure are significantly associated with stroke risk. We see that the probabilities of stroke in 10 years with these risk factors are higher than the risks for those without. Many of these variables change over time. In order to adjust for time-varying behavior of the covariates, we could no longer assume a constant hazard rate, used in Cox PH models. Options include implementing a time-dependent Cox model, or jointly modeling survival data and longitudinal data, capturing the variation in a given subject's measurements over time (Zhang 2018). This study is limited by the assumption that data is missing completely at random, the use of backward selection as a variable selection

Table 3: Change in covariates from Period 1 to Period 3

	Male	**Female**
Characteristic	**N = 1,102**[1]	**N = 1,438**[1]
Change in SYSBP	10 (18)	9 (18)
Change in BMI	0.00 (2.12)	0.28 (2.50)
Change in total cholesterol	-5 (38)	9 (40)
Change in BP medication use		
-1	3 (0.3%)	7 (0.5%)
0	987 (90%)	1,198 (83%)
1	112 (10%)	233 (16%)
CHD diagnosed after period 1	129 (12%)	90 (6.3%)
Diabetes diagnosed after period 1	76 (6.9%)	80 (5.6%)
Smoking Status Change		
-1	237 (22%)	155 (11%)
0	838 (76%)	1,263 (88%)
1	27 (2.5%)	20 (1.4%)

[1] Mean (SD); n (%)

procedure (not guaranteed to find the best model), and data availability for period 3.

Sources

Portet S. A primer on model selection using the Akaike Information Criterion. Infect Dis Model. 2020 Jan 7;5:111-128. doi: 10.1016/j.idm.2019.12.010. PMID: 31956740; PMCID: PMC6962709.

Zhang Z, Reinikainen J, Adeleke KA, Pieterse ME, Groothuis-Oudshoorn CGM. Time-varying covariates and coefficients in Cox regression models. Ann Transl Med. 2018 Apr;6(7):121. doi: 10.21037/atm.2018.02.12. PMID: 29955581; PMCID: PMC6015946.

Appendix

Schoenfeld Residual Plots

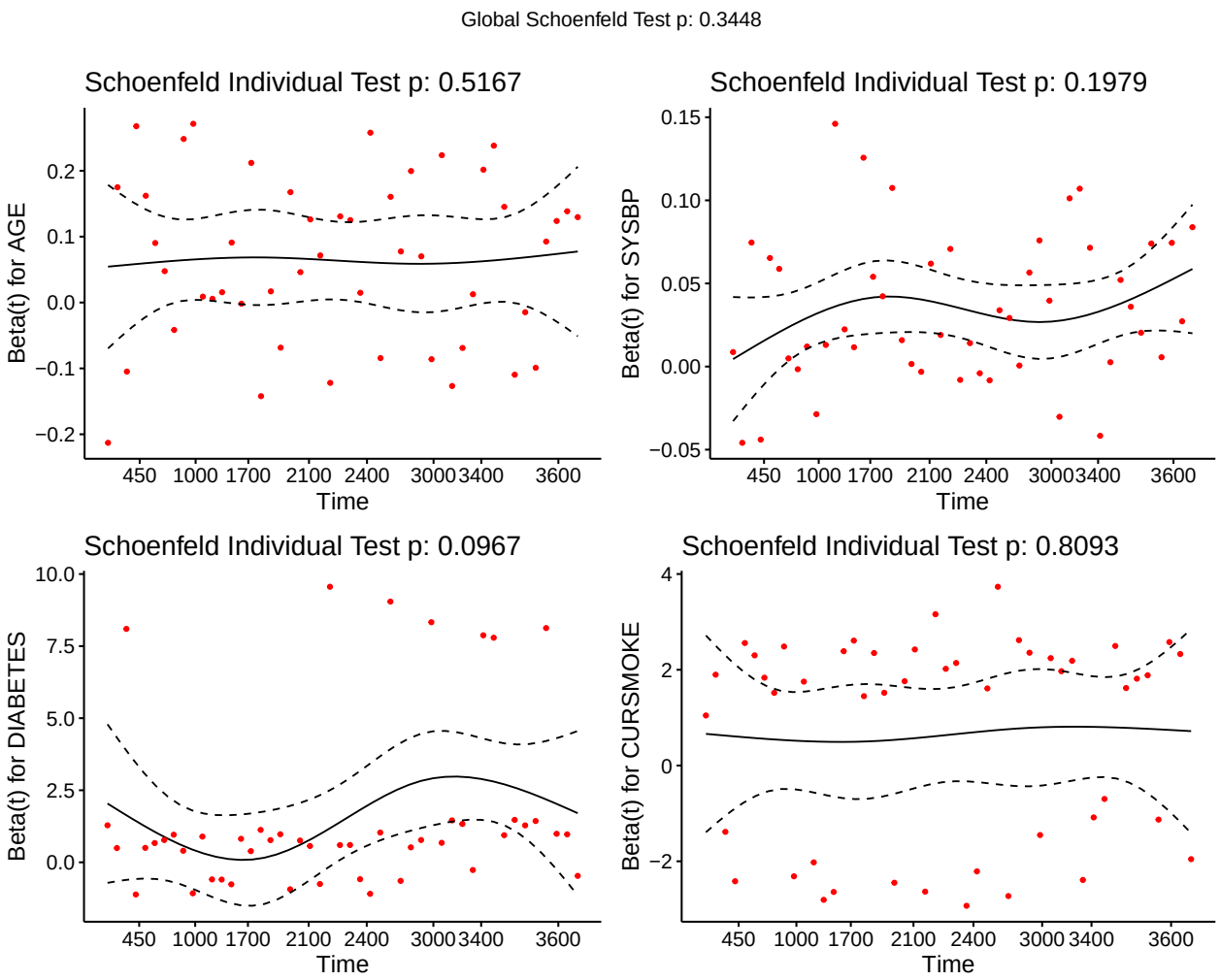

Figure 3: Schoenfeld Plots

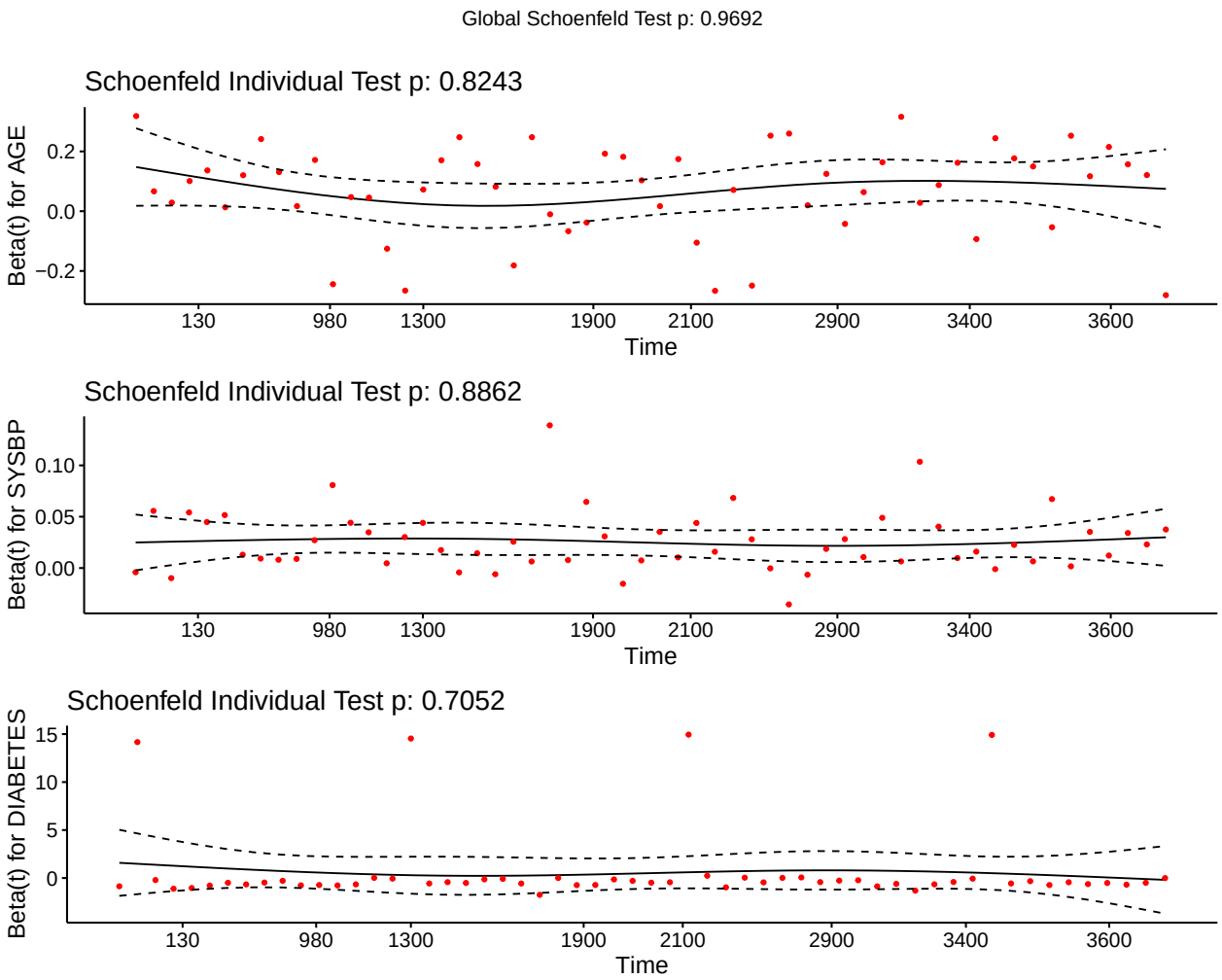

Figure 4: Schoenfeld Plots